Warm-Up Answer Key

Equations and Inequalities

LESSON 8-1 · UNDERSTAND EQUATIONS AND THEIR SOLUTIONS

MOVE *Check whether a value is a solution*
SLIDE 1) 20 2) 9 3) 4n = 20 4) no
MODEL 4 12 is

1. yes
2. no
3. yes
4. no
5. no
6. yes
7. 30
8. 3n = 15; yes

DROPPED *skill 3 – writing an equation from a sentence is this lesson's own second session, not something it assumes; item 8 bridges into it*

LESSON 8-2 · WRITE AND SOLVE EQUATIONS USING ADDITION OR SUBTRACTION

MOVE *Undo an addition or subtraction to find the unknown / Check by substituting the answer back in*
SLIDE 1) 5 2) 7.2 3) subtracting 8 4) 25
MODEL 11 36 36

1. x = 12; check: 12 + 7 = 19
2. n = 20; check: 20 − 6 = 14
3. a = 15; check: 15 + 25 = 40
4. b = 50; check: 50 − 18 = 32
5. c = 5.7; check: 5.7 + 3.5 = 9.2
6. d = 3/4; check: 3/4 − 1/4 = 1/2
7. 12
8. m − 14 = 9; m = 23

LESSON 8-3 · WRITE AND SOLVE EQUATIONS USING MULTIPLICATION OR DIVISION

MOVE *Undo a multiplication or division to find the unknown / Check by substituting the answer back in*
SLIDE 1) 56 2) 1.2 3) dividing by 4 4) 20
MODEL 6 7 7

1. x = 9; check: 4 × 9 = 36
2. x = 45; check: 45 ÷ 5 = 9
3. m = 8; check: 9 × 8 = 72
4. n = 42; check: 42 ÷ 7 = 6
5. p = 0.3; check: 12 × 0.3 = 3.6
6. w = 18; check: 1/3 × 18 = 6
7. 75
8. 8m = 96; m = 12

LESSON 8-4 · WRITE AND REPRESENT INEQUALITIES

MOVE *Turn a comparison phrase into an inequality symbol / Test a value against the inequality*
SLIDE 1) < 2) ≥ 3) 2 4) no
MODEL greater ≥ yes

1. ≥
2. ≤
3. ≤
4. > ; no
5. <
6. 42 inches
7. 8
8. p ≥ 4; yes

DROPPED *skill 3 – placing a point and shading a number line is this lesson's own second session and all of 8-5; on the worksheet the boundary number stands in for the picture*

MOVE *Read the boundary, the circle, and the direction from the symbol / Test a value against the boundary*
SLIDE 1) left 2) closed 3) no 4) 10
MODEL 3 closed right

1. 5; closed; right

2. 2; open; left

3. 8; closed; left

4. 0; open; right

5. yes

6. no

7. $x \leq 12$

8. 4 and 5